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Research paper

Fluid flow in UV nanoimprint lithography with patterned templates

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ABSTRACT

Fluid flow in UV nanoimprint lithography with patterned template is simulated to study the effect of pattern directionality on droplet merging, throughput and defectivity. Flow in high density patterns is simulated by using an anisotropic permeability. A method to calculate the permeability for templates with line and space patterns is presented. It is found that the directionality of the pattern results in anisotropic spreading of the droplets. The droplets flow faster in the direction of the pattern resulting in channelized unfilled regions. These regions fill slowly because the permeability in the direction normal to the patterns is low. A modified hexagonal arrangement for droplet dispensing is proposed in which the filling time is lower than square and hexagonal droplet arrangements. Simulations are carried out with different multi-patterned template and droplet dispensing schemes. Dispensing droplets such that their volume is sufficient to fill the features locally is found to be the optimum dispensing scheme to minimize filling time.

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1. Introduction

UV nanoimprint lithography (UVNIL) is a pattern replication process used to transfer micro- and nano- patterns from templates onto glass or silicon substrates for high volume manufacturing [1] of display applications [2–4], photonic devices [5,6], photovoltaics [7] and high density bit pattern media [8,9]. A typical set up for UVNIL is shown in Fig. 1. An inkjet dispenser is used to dispense droplets of low viscosity photo-curable imprint material on the substrate. These droplets are a few picoliters in volume. A template with micro- or nano- features is lowered on the substrate. As the template lowers, the droplets merge together and fill the gap between the template and the substrate to form a uniform film. Typically, the final gap is sub-100 nm. When the desired gap is reached, UV light irradiates the imprint film, which photo-polymerizes to form a solid resist layer with features replicated from the template. The template is then peeled from the resist with a controlled mechanism completing the patterning process.

High throughput and low defectivity in UVNIL are important to achieve high volume manufacturing for applications such as flash memory. These factors depend strongly on the template control scheme [10,11], droplet distribution [11] and template pattern [11–13]. The effect of template control scheme and droplet distribution on throughput of UVNIL has been explored previously using modeling and simulation [10,11,14,15]. The patterned templates used in UVNIL are very high quality as defects on template patterns can result in repeating defects on the final pattern. They are fabricated from an industry standard mask blank using phase shift mask fabrication technology [16,17]. The template fabrication process is highly precise and contributes considerably to the over-all cost of ownership [16,18]. The throughput and defect rate is influenced by the feature size and pattern density. Since the cost of template fabrication is very high, experimenting with different template designs to minimize filling time is not viable. We can minimize filling time by understanding the fluid flow behavior for various pattern type and sizes through modeling and simulation. The effect of patterns on fluid flow has been studied primarily in the context of feature filling and gas trapping [19–25]. The directionality of pattern features has a huge influence on the fluid flow dynamics during spreading of the droplets [13]; however, the bulk effect of high density patterns on droplet spreading across the template has not been explored. Simulating the flow around individual patterns is computationally very expensive due to the high density of patterns which creates a huge bottleneck in studying the fluid behavior.

Here we propose a model for droplet spreading in UVNIL with patterned template which incorporates the effect of patterns on fluid flow by calculating an average flow permeability of the template. UVNIL with line and space patterned templates is studied by varying droplet size and arrangement. The average permeability accounts for the porosity due to anisotropic patterns on the imprint template. Droplet spreading is simulated with droplets dispensed in different arrangements, multi-patterned templates and different droplet dispensing

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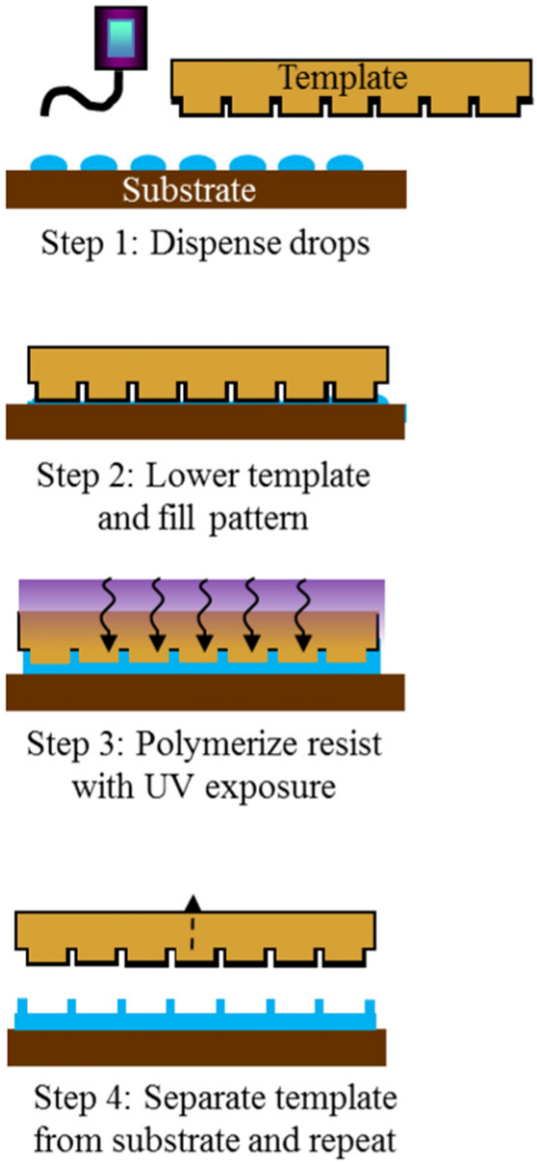


Fig. 1. Main steps for pattern replication in UV Nanoimprint lithography.

schemes to find the optimum droplet distribution for maximum throughput. We identify and characterize different non-fill defects formed when using different patterns and predict the size, location and count of these defects.

2. Simulation method

A schematic of imprint droplets between a patterned template and a substrate is shown in Fig. 2. The template can be controlled to move with a constant velocity, constant pressure or applied force [10]. As the template approaches the substrate, the liquid-air interface of the droplet results in an attractive capillary force on the template. The fluid flow in the gap results in viscous force which balances the capillary force due to the liquid interface and any externally applied force.

Typically sub-ten picoliter droplets are used in UVNIL [26]. The initial radii of the droplets are about 10–100 μm depending on their size and the wettability of the substrate. For sub-100 nm wide patterns, the droplets are many orders of magnitude larger than the patterns and each droplet cover hundreds or thousands of individual patterns. Since the resist flows over a very high density of patterns, the resistance to flow can be described as an effective permeability. The permeability **K**

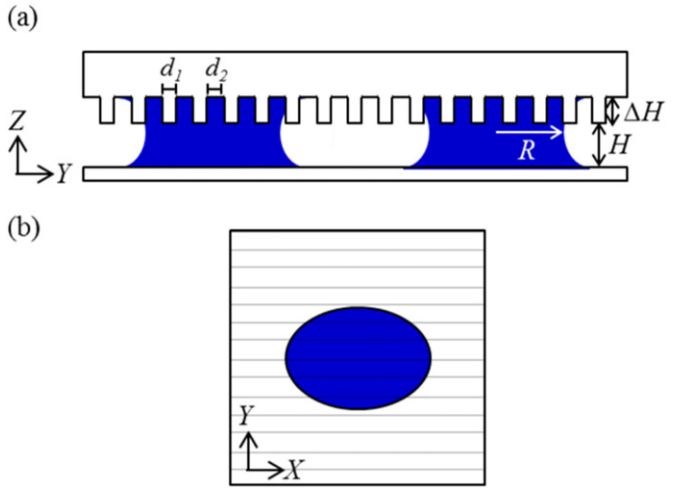


Fig. 2. Schematic of droplets between a substrate and a template with line and space patterns. (a) Side view of the template. The feature height of the pattern is ΔH. The minimum gap between the substrate and the template is H. The width of the line and space pattern is d_1 and d_2 respectively. The patterns are parallel to the x direction. (b) Top view of the template with a droplet spreading under it. The horizontal lines show the direction of the patterns (not to scale).

is in general a tensor and anisotropic reflecting the nature of the pattern. The equations describing the pressure and velocity fields in the drop for UVNIL with high density patterned template are given by the Reynolds' lubrication theory:

$$\frac{\partial H}{\partial T} = \nabla \cdot \left(\mathbf{K} \cdot \frac{\nabla P}{\mu} H \right), \tag{1}$$

$$\mathbf{U} = -\mathbf{K} \cdot \frac{\nabla P}{\mu} \tag{2}$$

where H is the gap between substrate and template, P is the pressure in the droplet, $\mathbf{U}$ is the vertically averaged fluid velocity and μ is the viscosity of the imprint resist. For a rigid substrate and template, $\partial H/\partial T = -V$, where V is the template velocity. The pressure distribution in the fluid is calculated using Eq. (1) and the pressure boundary condition at the liquid-air interface. For a patterned template, the capillary pressure at the interface, $P_{interface}$ can be written as.

$$P_{\,interface} = P_{atm} - \frac{\gamma(\cos\theta_1 + \cos\theta_2)}{\hat{H}}, \tag{3}$$

where $\hat{H}$ is an averaged gap between the substrate and the template accounting for the effect of pattern roughness on the capillary pressure. An expression of $\hat{H}$ for line and space pattern is presented in the Supplementary information. P_{atm} is the atmospheric pressure and γ is the surface tension of the imprint resist. θ_1 and θ_2 are contact angles made by the imprint material with the template and the substrate respectively. Imprint material used in UVNIL is highly wetting and for our simulations we assume that $\theta_1 = \theta_2 = 0$, so, $P_{\,interface} = P_{atm} - 2\gamma/\hat{H}$. In our model, we assume that UVNIL is being carried out in vacuum or in a gas which is highly soluble in imprint material and so gas release and trapping can be neglected. The imprint process begins with the droplets making contact with both the substrate and the template. We also assume that the feature filling is instantaneous as the feature fill time is negligible compared to the time required for the fluid to spread in the gap.

The governing equations and boundary condition [Eq. (1), (2) and (3)] can be non-dimensionalized using the characteristic values of pressure, gap thickness, time and velocity. These values are given by: $P_c = 2\gamma/H_o, H_c = H_o, T_c = 6\mu L^2/\gamma H_o$, $V_c = \gamma H_o^2/6\mu L^2$ and $U_c = \gamma H_o/6\mu L$, where H_o is the initial gap. The x and y coordinates are non-

dimensionalized using the width of the template L. The dimensionless governing equations are given by

$$-v = \nabla \cdot ((\mathbf{k} \cdot \nabla p)h), \tag{4}$$

and

$$\mathbf{u} = -\mathbf{k} \cdot \nabla p, \tag{5}$$

where, $h = H/H_c$, $t = T/T_c$, $v = V/V_c = \partial h/\partial t$, $p = P/P_c$, and $\mathbf{u} = \mathbf{U}/U_c$. v is the dimensionless template velocity. The dimensionless pressure boundary condition is given by

$$p_{\,interface} = -\frac{1}{\hat{h}}, \tag{6}$$

where $\hat{h}$ is the dimensionless averaged gap between the substrate and the template accounting for the effect of pattern roughness on the capillary pressure.

A schematic of a nanoimprint template with line and space patterns is shown in Fig. 2. The feature height of the pattern is ΔH and the minimum and maximum gap between the substrate and the template at any time is H and $H + \Delta H$ respectively. The width of lines and spaces is d_1 and d_2 respectively. The lines and spaces in Fig. 2(a) are parallel to the X direction and into the plane. The resist completely fills the features and the gap between the template and the substrate at the end of the imprinting process to form a continuous resist layer. The total volume of resist Q required to fill the gap is given in dimensional form by

$$Q = L^2\left(H_f + \Delta H\frac{d}{1+d}\right), \tag{7}$$

where, H_f is the final resist thickness and $d = d_2/d_1$. Typically the pattern height ΔH is less than 100 nm.

The dimensionless governing Eqs. (4) and (5) can be written in the Cartesian coordinate system as

$$-v = \frac{\partial}{\partial x}\left(k_{xx}h_{avg}\frac{\partial p}{\partial x}\right) + \frac{\partial}{\partial y}\left(k_{yy}h_{avg}\frac{\partial p}{\partial y}\right), \tag{8}$$

$$u_x = -k_{xx}\nabla p, \tag{9}$$

and

$$u_y = -k_{yy}\nabla p. \tag{10}$$

For line and space patterns, h_{avg}, k_{xx} and k_{yy} are given by

$$h_{avg} = h\left(\frac{1 + (1 + \Delta h/h)d}{1+d}\right), \tag{11}$$

$$k_{xx} = h^2\frac{1 + (1 + \Delta h/h)^3 d}{1 + d(1 + \Delta h/h)}, \tag{12}$$

and

$$k_{yy} = h^2\left(\frac{1+d}{1 + (1 + \Delta h/h)^{-3}d}\right)\left(\frac{1+d}{1 + (1 + \Delta h/h)d}\right), \tag{13}$$

where $d = d_2/d_1$ and $\Delta h = \Delta H/H_c$. Fig. 3 shows the plot between permeability k_{xx} and k_{yy} in the x and y direction as the gap closes from $h = 1$ to $h = 0.01$ for $\Delta h = 0$, 0.01 and 0.1. For $\Delta h = 0$ (pattern-free template), both k_{xx} and k_{yy} scale as h^2. For line and space patterns, k_{xx} becomes much larger than k_{yy} as the gap closes.

The dimensionless pressure boundary condition is given by

$$p_{\,interface} = -\frac{1}{h}\frac{1}{1+d}\left(1 + \frac{d}{1 + (\Delta h/h)}\right). \tag{14}$$

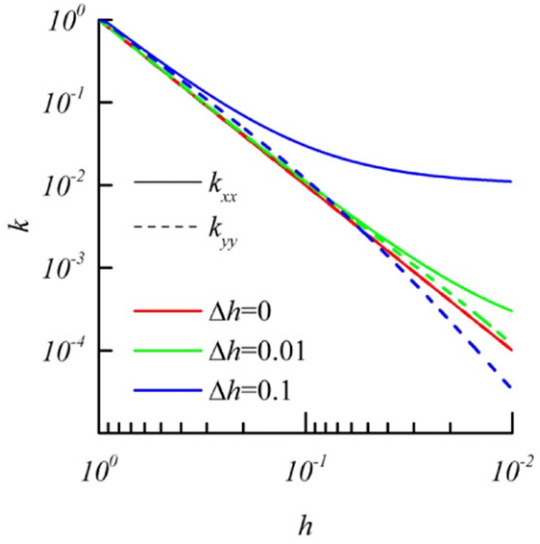


Fig. 3. Permeability k_{xx} and k_{yy} in the x and y direction respectively for pattern height $\Delta h = 0$, 0.01 and 0.1. k_{xx} and k_{yy} are equal for pattern-free template and scale as h^2. For $\Delta h = 0.1$, the permeability becomes significantly higher in the x direction compared to y direction as the gap closes.

The derivation of these expressions is shown in the Supplementary information. The dimensionless volume of resist required to fill entire gap at $h = h_f$ is given by

$$q = h_f + \Delta h\frac{d}{1+d}. \tag{15}$$

q increases as Δh increases. A method to solve these equations and simulating droplet spreading using the Volume of Fluid (VOF) method [27] has been described previously [11].

3. Results

3.1. UVNIL using template with line and space patterns

Simulation of UVNIL using a template with line and space patterns with nine droplets and 100 droplets dispensed in a square arrangement is shown in Figs. 4 and 5, respectively. The droplet spreading and merging is simulated as the gap between the substrate and the template closes from an initial gap of $h = 1$ to a final gap of $h = 0.01$. The template has zero net force acting on it and the feature height $\Delta h = 0.1$. For initial gap $H_o = 1$ μm, $h = 1$ and $\Delta h = 0.1$ corresponds to a dimensional gap of $H = 1$ μm and a feature height of $\Delta H = 100$ nm respectively as shown in Fig. 3(a). For a pattern width 100 nm and substrate width 10 cm, each drop spans about 10,000 gratings for nine drops and 3000 gratings for 100 drops. The width of lines and spaces are equal i.e. $d = 1$. The patterns are parallel to the x-direction. The initial volume of the droplet is selected based on the number of droplets and Eq. (10) so that the resist completely fills the features forming a uniform continuous film at $h = 0.01$.

We find that the droplets are stretched when UVNIL is carried out with templates having line and space patterns. The non-circular shape is a result of the difference in flow permeability in x and y directions. Fig. 3 shows that for $\Delta h = 0.1$, permeability k_{xx} in the x-direction becomes significantly higher than k_{yy} as the gap becomes smaller. This leads to the droplet spreading faster in the x direction compared to the y direction. The spreading of the droplets can be divided into three regimes: (i) initial circular spreading (ii) elliptical spreading before droplet merging and (ii) linear spreading after droplet merging. Initially from $h = 1$ to about $h = 0.6$, the droplets spread uniformly in the all directions as the droplets are squeezed by the template. This occurs

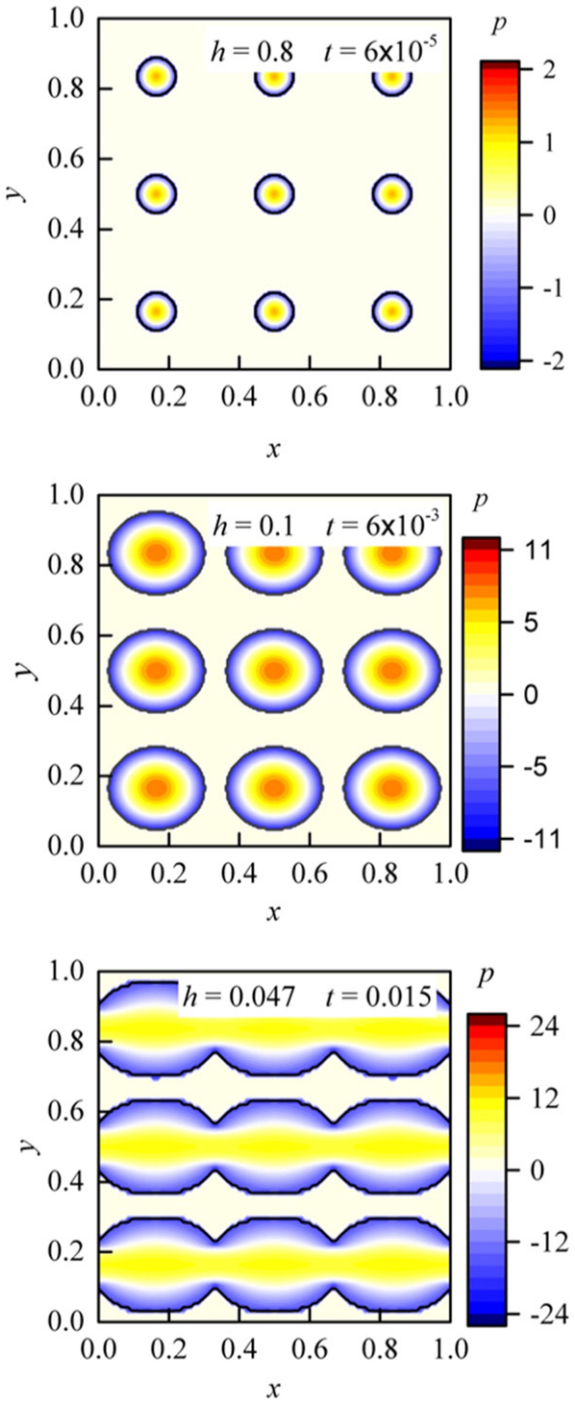


Fig. 4. Contour map showing pressure field for nine drops spreading on a substrate at $h = 0.8$, $h = 0.1$ and $h = 0.047$. The template has line and space patterns and has net zero force acting on it. The grey horizontal lines show the direction of the patterns (not to scale). The droplet spreads faster in the x direction compared to y direction. The feature height Δh is 0.1.

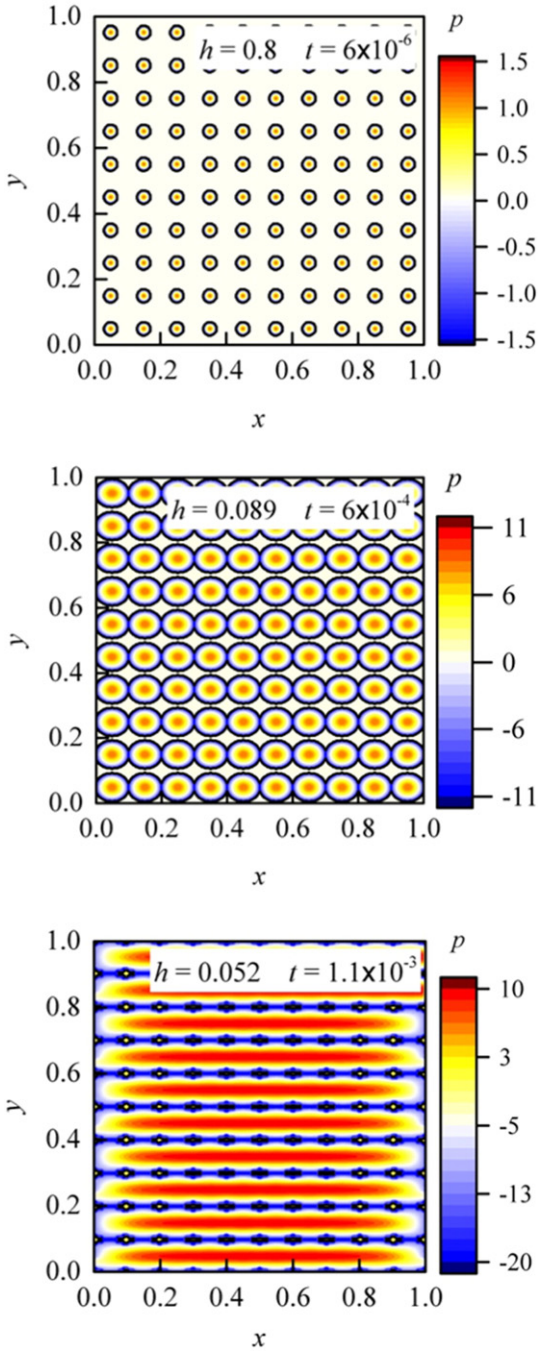


Fig. 5. Contour map showing pressure field for 100 drops spreading on a substrate at $h = 0.8$, $h = 0.089$ and $h = 0.052$. The template has line and space patterns and has net zero force acting on it. The grey horizontal lines show the direction of the line and space patterns (not to scale). The droplet spreads faster in the x direction compared to y direction. The feature height Δh is 0.1.

because the permeabilities in x- and y- directions are about the same as shown in Fig. 3. In this regime both k_{xx} and k_{yy} scale as $O(h^2)$. Thus the droplets maintain their circular shapes as shown in Figs. 4 and 5 (top). As the droplets are squeezed further, k_{xx} starts to increase rapidly while k_{yy} continues scaling with h^2. Thus the droplet spreads faster in the x direction compared to the y direction leading to droplets of ellipse like shape as shown in Figs. 4 and 5 (center). Droplets continue spreading under this regime until they either start merging with neighboring droplets or reach the template edge. Once all the droplets in x directions merge together, they form channels flowing only in the y direction as shown in Figs. 4 and 5 (bottom). Since k_{yy} is very small compared to k_{xx}, the fluid spreads slowly in the y direction after drop merging.

The droplet merging simulation is repeated with multiple droplets in a square arrangement and three pattern heights $\Delta h = 0$, 0.01 and 0.1. The dimensionless gap height h as a function of dimensionless filling time t is shown for multiple droplets in Fig. 6. The results for $\Delta h = 0$ corresponds to results for a template with no patterns. The filling time for $\Delta h = 0$ reduces as the number of droplets increases. The filling time required for $\Delta h = 0.01$ does not differ significantly from $\Delta h = 0$ because there is a relatively small difference in permeability for $\Delta h = 0$ and Δh

$= 0.01$ as shown in Fig. 3. However, a pattern height of $\Delta h = 0.1$ has significant impact on the filling time. The filling time is longer for $\Delta h = 0.1$ compared to $\Delta h = 0$ because of the higher volume of the resist required to fill a gap with $\Delta h = 0.1$. A change in gradient is observed in the plot for $\Delta h = 0.1$. This is because of slow droplet spreading once the droplets merge in the x direction to form channels as shown in the Figs. 4 and 5 (bottom). The characteristic time for these simulation is $T_c = 6\mu L^2/\gamma H_o$. For UVNIL with substrate length $L = 10$ cm, initial gap height of $H_o = 1$ μm, resist viscosity $\mu = 0.001$ Pa.s and surface tension $\gamma = 70$ dyne/cm, $T_c = 857$ s. Fig. 6 shows that without pattern, the overall filling time is about 200 s for nine droplets and 44 s for 100 droplets. With a pattern height of 10 nm, the overall filling time is about 220 s for 9 drops and about 48 s for 100 droplets. For larger patterns of height 100 nm, the overall filling time increases to about 370 s for nine droplets and 80 s for 100 droplets. Fig. 6(b) shows that the results for multiple droplets collapse into one curve when filling time is rescaled by the number of drops N suggesting that t scales as $1/N$.

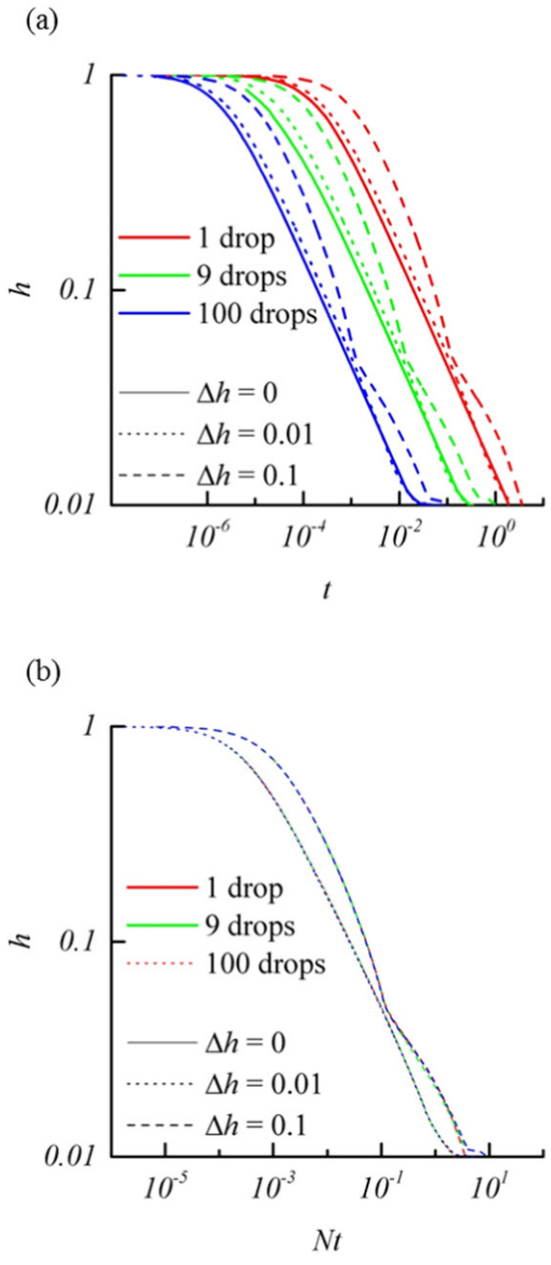


Fig. 6. (a) Filling time for UVNIL with multiple droplets and a template with line and space patterns of height $\Delta h = 0$, 0.01 and 0.1. The droplets are dispensed in a square arrangement. (b) Rescaling the filling time with the number of droplets (N) collapses the results into one curve suggesting that t scales as N^{-1}

The overall filling time also depends on the droplet arrangement [28]. The droplet spreading simulation is repeated with 63 droplets and 101 droplets dispensed in a hexagonal arrangement. Fig. 8(a) shows the droplets dispensed in a hexagonal arrangement that $l_h/l_b \approx \sqrt{3}/2 (= 0.87)$. Fig. 9(b) shows these droplets spreading on the substrate. The hexagonal droplet arrangement results in unfilled regions at the corner which require more time to fill. The dimensionless gap height h as a function of dimensionless filling time t is shown in Fig. 9. The filling time increases as the pattern height is increased. For the same number of droplets, the filling time is found to be higher in hexagonal arrangement compared to square arrangement.

We propose a modified hexagonal arrangement in which more droplets are added at the corner and the distance between them is increased to cover the corner and edges of the substrate. Fig. 7(b) shows droplets arranged in modified hexagonal arrangement with $l_h^*/l_b^* = 0.94$. Fig. 8(c) shows the droplets spreading on the substrate. The gap height as a function of filling time for about 100 droplets in square, hexagonal and modified hexagonal arrangement is shown in Fig. 10. The figure shows that the filling time is shortest when the droplets are arranged in modified hexagonal arrangement.

These results from simulations can be compared with the NIL experiments performed with patterned templates. Zhang et al. performed NIL with templates patterned with 20 nm wide arrays on a 26 mm × 33 m field [29]. These experiments confirm faster filling of the resist in the direction parallel to the patterns resulting in longer filling times as shown in Fig. 9 and formation of channels similar to the simulations shown in Fig. 8. The formation of channels has also been observed by Fletcher et al. who performed NIL with patterned templates with 32 nm gratings and droplets in a square arrangement [30]. Here we have shown that the filling time can be reduced by arranging the droplets in a modified

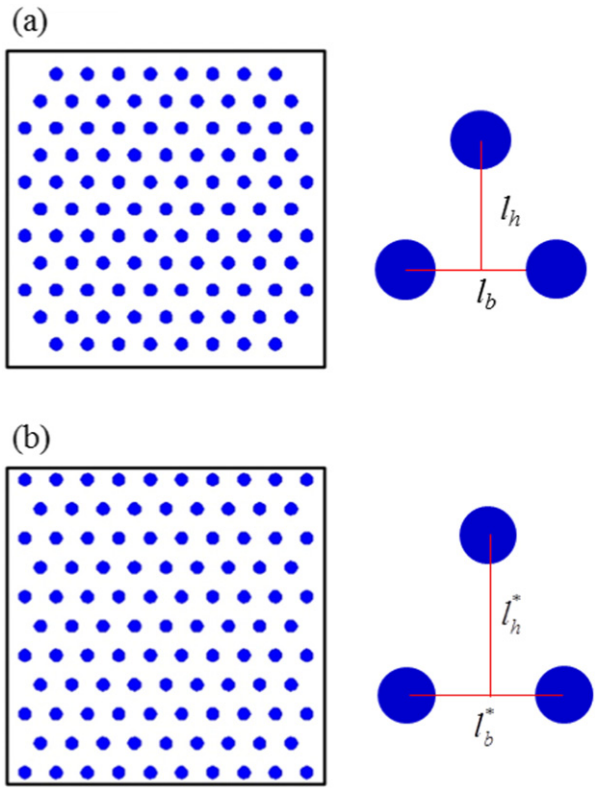


Fig. 7. Schematic showing top view of droplets dispensed on a substrate in two arrangements (a) hexagonal and (b) modified hexagonal arrangement. The value of l_h/l_b and l_h^*/l_b^* is 0.87 and 0.94, respectively.

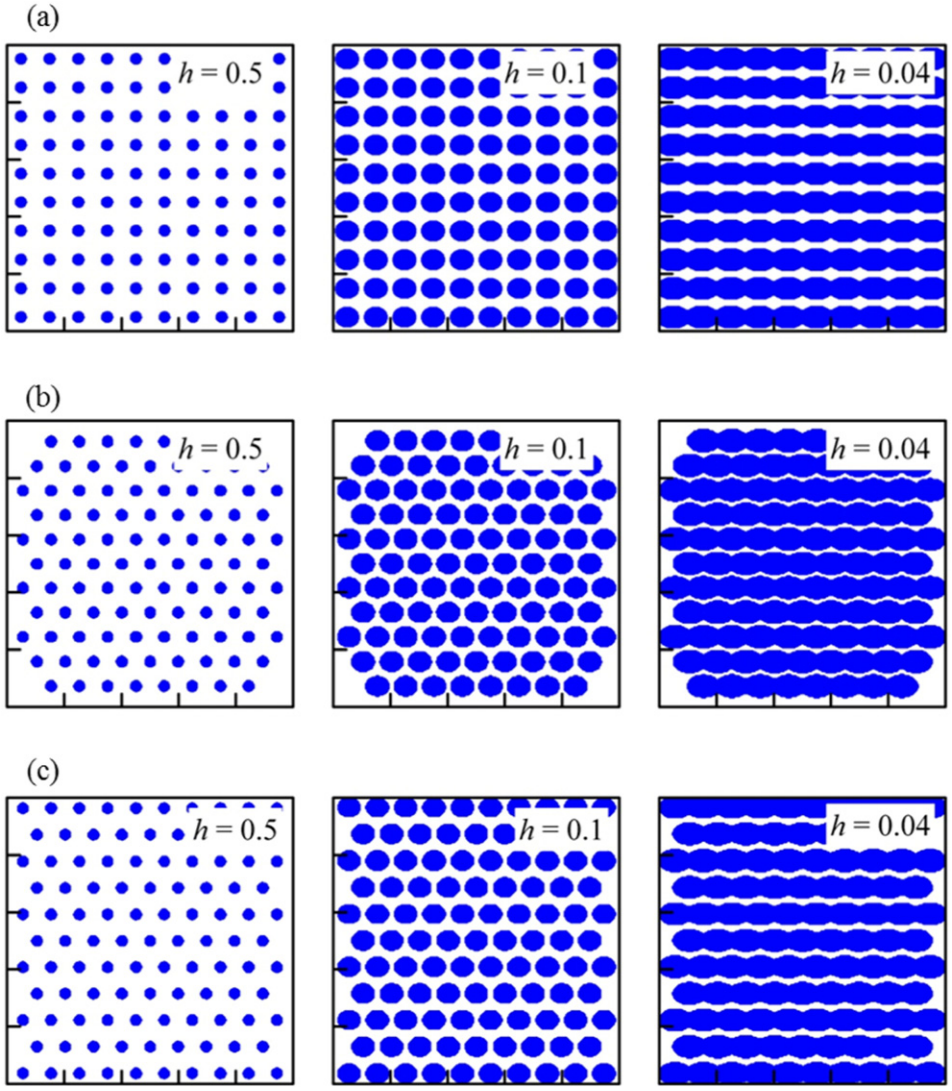


Fig. 8. Simulation of UVNIL with about 100 droplets dispensed in (a) square (b) hexagonal and (c) modified hexagonal arrangement. The gap is completely filled with resist at $h = 0.01$.

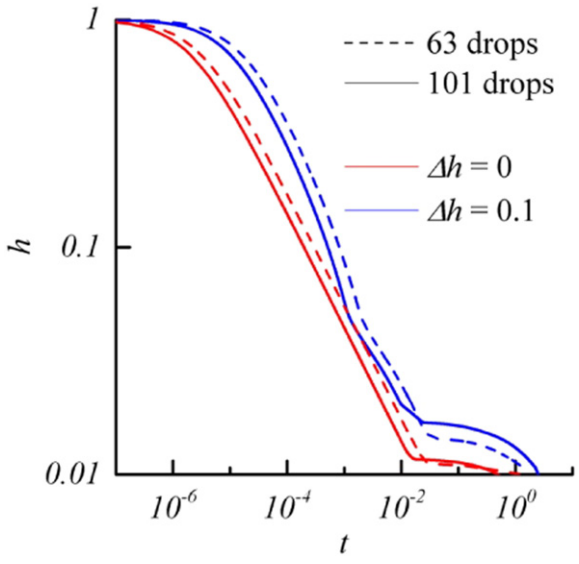


Fig. 9. Filling time for UVNIL with multiple droplets and a template with line and space pattern for pattern height $\Delta h = 0$, 0.01 and 0.1. The droplets are dispensed in a hexagonal arrangement.

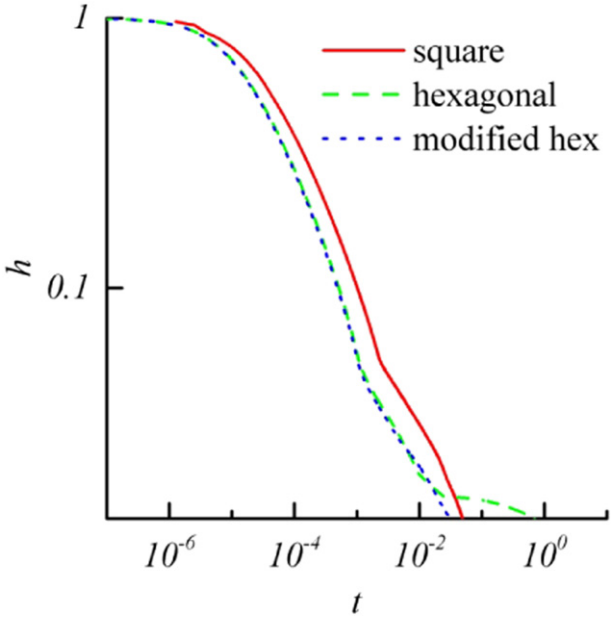


Fig. 10. The gap height h as a function of filling time t for droplets dispensed in square (100 drops, solid), hexagonal (101 drops, dash) and modified hexagonal (105 drops, dot) arrangements. The overall filling time is least for droplets dispensed in a modified hexagonal arrangement.

hexagonal arrangement. Experiments performed by Zhang et al. confirm that the filling time can be optimized by arranging the droplets in a "diamond-like" drop pattern and optimizing distance between the droplets [29].

3.2. UVNIL using multi-patterned template

Multi-patterned templates have varying flow permeability in different parts of the template which can lead to interesting fluid flow behavior. We have simulated drop spreading on three types of templates which are shown in Fig. 10. In template type-I, the top half (section A) is covered with line and space patterns of height ΔH parallel to x direction and the bottom half (section B) is pattern free. In template type-II, section A is covered with lines and space patterns of height ΔH parallel to y direction and section B is pattern free. In template type-III, both section A and section B are pattern free however section A is at a height $\Delta H/2$ higher than section B as shown in Fig. 10(c). At a fixed gap height H, same volume of resist is required to fill the gap between the template and the substrate for all three templates.

The permeability in sections A and B are different resulting in different flow behaviors as was shown in Fig. 3. The spreading and merging is simulated with these multi-patterned templates as the gap between the substrate and the template closes from an initial gap of $h = 1$ to a final gap of $h = 0.01$. The template has no net force acting on it. $h = 1$ and corresponds to 1 μm. The width of lines and spaces are equal i.e. $d = 1$. Here we explore two drop pattern schemes. In the first drop pattern scheme, all the droplets have equal volume and the total volume of droplets is sufficient to fill the entire template when $h = h_f$. In the

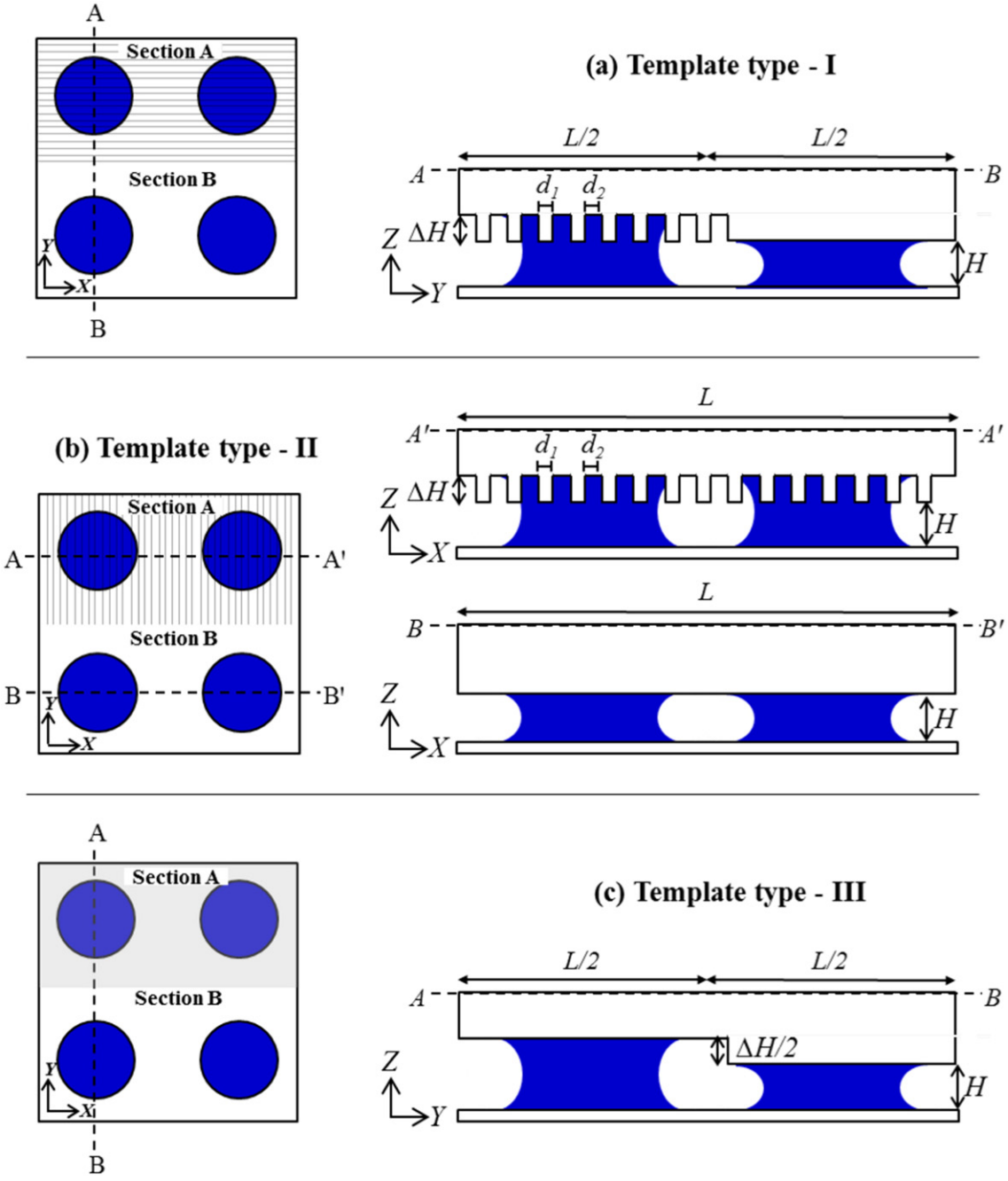


Fig. 11. Schematic showing top view and cross-sectional view of multi-patterned template type- I, II and III. Template type-I (a) and type-II (b) have line and space patterns at the top (section A) and no pattern at the bottom (section B). The grey lines show the direction of the patterns (not to scale). The patterns are parallel to x and y direction in template I and II respectively. Template type-III does not have patterns but the gap in section A is larger than section B by $\Delta H/2$.

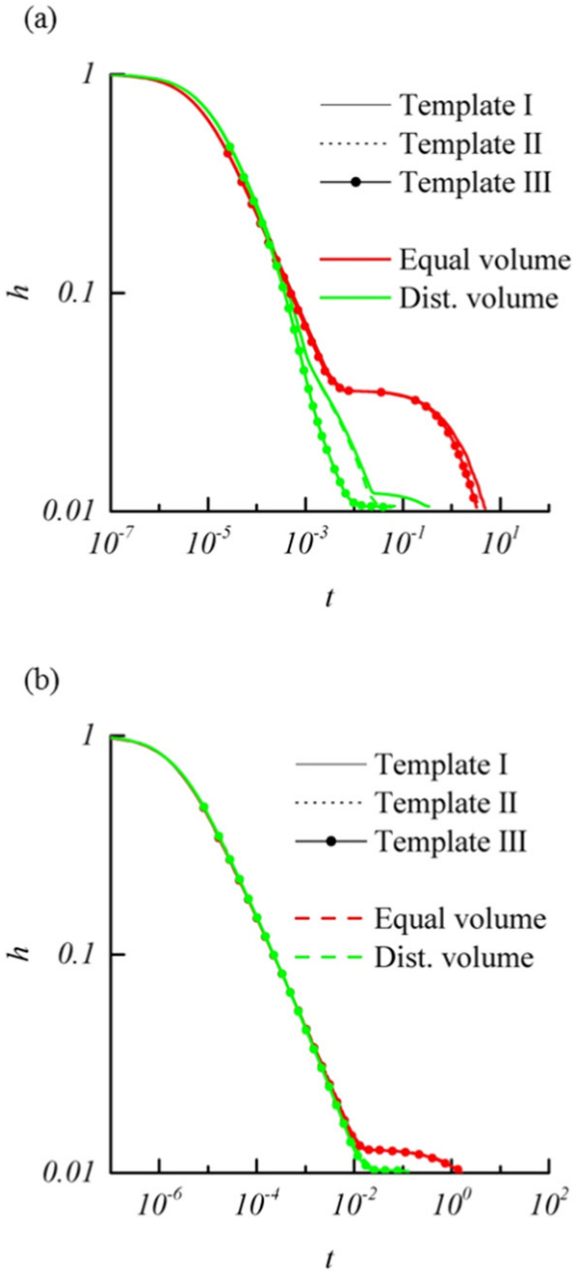


Fig. 12. Filling time for UVNIL with 100 droplets, feature height $\Delta h = 0.1$ and templates I, II and III. Droplets are dispensed under two schemes: equal drop volume and distributed drop volume. The filling time for equal drops is found to be much longer than distributed drop volume.

second drop pattern scheme, the droplet volume is distributed such that each section has different size of droplets such that total volume of the droplets in a section is sufficient to completely fill that section when $h = h_f$. In both schemes, the initial volume of the droplet is selected such that the resist completely fills the features and the entire substrate forming a uniform continuous film at $h = 0.01$. The total number of drops in each section is $N/2$. At the beginning of UVNIL, the dimensionless height of droplets in section A and B is $(1 + 0.5\Delta h)$ and 1 respectively. The dimensionless radius of droplets used for the two sections in the two schemes is shown in Table 1.

The filling time for template I, II and III under the two drop pattern schemes for 100 drops and feature height is $\Delta h = 0.1$ is shown in Fig. 12(a). The filling time using equal drop volume is higher compared to distributed drop volume for all three templates. The total number of droplets in each section on the template is $N/2$. We can calculate the total volume of droplets in each section using Table 1. For equal drops, the total volume of droplets in each section is $0.5(h_f + 0.25\Delta h)$. The volume required to fill section A is $0.5(h_f + 0.5\Delta h)$. Thus there is a deficit of $0.125\Delta h$ in section A and excess volume of same amount in section B. This excess fluid flows from section B to section A as the gap between the template and the substrate closes which increases the overall filling time. The results for the different templates under the same scheme are comparable.

For distributed drop volume, the total volume of droplets in section A and B is $0.5(h_f + 0.5\Delta h)$ and $0.5h_f$ respectively. The volume required to fill section A and B is $0.5(h_f + 0.5\Delta h)$ and $0.5h_f$ respectively. Thus unlike equal drops, distributed drops provide sufficient volume of fluid in each section leading to significantly shorter filling times. These simulations were repeated with same type of templates but having pattern height $\Delta h = 0.01$. The filling time as a function of gap height is shown in Fig. 12(b). Distributed drop volume scheme still provides lower filling time than dispensing drops equally. Based on these results it can be concluded that irrespective of pattern direction in multi-patterned templates, minimum filling time is provided by dispensing drops such the volume is distributed based on local filling requirement. The difference in filling time increases with increase in the pattern height.

4. Conclusions

A model for UVNIL with a patterned template is presented to simulate droplet spreading and analyze the defectivity. An average permeability is calculated to take into account the effect of fluid flow around features. For line and space patterns, the droplet spread faster in the direction parallel to the patterns due to increase in permeability while the flow in the direction normal to the patterns is slow leading to intermediate elliptical droplets and unfilled channels. The flow anisotropy and overall filling time increase with the pattern height, which are confirmed by experiments in the literature. Droplet spreading is simulated for droplets dispensed in square, hexagonal and modified hexagonal arrangements and it is found that the filling time is shortest for droplets dispensed in the proposed modified hexagonal arrangement. Simulation is carried out for UVNIL with templates which are pattern free in one half and are covered with line and space patterns on the other half. Two drop pattern schemes are explored for UVNIL with these multi-patterned templates to optimize the process for minimum filling time. It is found the dispensing the drops such that the droplets have sufficient volume to fill the gap locally leads to shorter filling times compared to dispensing equal volumes of droplets on the entire substrate. This computational tool can help make better design decisions for creating templates and drop dispensing strategies that improve throughput and defect rate.

Table 1
Size of droplets on section A and B for two droplet schemes in Fig. 11: equal drop volume and distributed drop volume.

Section of Template I,II or III	Volume required to fill section	Droplet radius for Scheme 1 (equal drop volume)	Droplet radius for Scheme 2 (distributed drop volume)
Section A	$0.5(h_f+0.5\Delta h)$	$\left[\frac{h_f+0.25\Delta h}{N\pi(1+0.5\Delta h)}\right]^{1/2}$	$\left[\frac{h_f+0.5\Delta h}{N\pi(1+0.5\Delta h)}\right]^{1/2}$
Section B	$0.5h_f$	$\left(\frac{h_f+0.25\Delta h}{N\pi}\right)^{1/2}$	$\left(\frac{h_f}{N\pi}\right)^{1/2}$

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Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.mee.2017.04.001.

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